```
import pandas as pd ## Pandas is our main data cleaning library
import numpy as np ## Numpy is our main numerical library
import matplotlib.pyplot as plt ## Matplotlib is our main plotting library
import seaborn as sns ## Seaborn is a high level plotting library
sns.set()
```

Load Data

```
kirb21_df = pd.read_csv("https://raw.githubusercontent.com/smart-stats/ds4bio_book/main/book/assetts/kirby21AllLevels.csv")
```

Filter to only inlcude one observation

```
#print(kirb21_df.tail())
kirby906a_ax_df = kirb21_df[kirb21_df["rawid"] == "kirby906a_ax.img"]
kirby906a_ax_df.head()
```

	Unnamed: 0	rawid	roi	volume	min	max	mean	std	type	level	id	icv	tbv
12540	12541	kirby906a_ax.img	Telencephalon_L	467063	2.0	350.0	165.2599	57.1707	1	1	906	1195015	1123076
12541	12542	kirby906a_ax.img	Telencephalon_R	470488	2.0	337.0	171.8695	59.3001	1	1	906	1195015	1123076
12542	12543	kirby906a_ax.img	Diencephalon_L	8801	60.0	327.0	227.1878	31.2303	1	1	906	1195015	1123076
12543	12544	kirby906a_ax.img	Diencephalon_R	9054	63.0	415.0	231.6770	31.1780	1	1	906	1195015	1123076
12544	12545	kirby906a_ax.img	Mesencephalon	9564	86.0	352.0	269.1003	28.6454	1	1	906	1195015	1123076

Next steps: Generate code with kirby906a_ax_df New interactive sheet

1. Write code to calculate this subject's volume of their type I level I telencephalon (left and right). Name the variables "telencephalon_L_volume" and "telencephalon_R_volume" such that when printed only the volume is printed (do not make it a dataframe, series, etc. only a number!). Also, find and print out the fractions of the ICV and TBV made up by their telencelphalon.
2. Create a dataframe that has the sum of all of the regions for every combination of type and level, name it "all_regions_df". So, it should have three columns, type, level and total_volume.

```
kirby906a_ax_df = kirby906a_ax_df.assign(telencephalon_L_volume = kirby906a_ax_df[kirby906a_ax_df["roi"] == "Telencephalon_L"]["volume"].item())
kirby906a_ax_df = kirby906a_ax_df.assign(telencephalon_R_volume = kirby906a_ax_df[kirby906a_ax_df["roi"] == "Telencephalon_R"]["volume"].item())
print("#this is the Left telencephalon volume, proportion of ICV, and proportion of TBV")
print(kirby906a_ax_df["telencephalon_L_volume"].iloc[0])
# index using ".iloc[]"
print(kirby906a_ax_df["telencephalon_L_volume"].iloc[0]/kirby906a_ax_df["tbv"].iloc[0])
print(kirby906a_ax_df["telencephalon_L_volume"].iloc[0]/kirby906a_ax_df["icv"].iloc[0])

print("#this is the Right telencephalon volume, proportion of ICV, and proportion of TBV")
print(kirby906a_ax_df["telencephalon_R_volume"].iloc[0])
print(kirby906a_ax_df["telencephalon_R_volume"].iloc[0]/kirby906a_ax_df["tbv"].iloc[0])
print(kirby906a_ax_df["telencephalon_R_volume"].iloc[0]/kirby906a_ax_df["icv"].iloc[0])
```

```
#this is the Left telencephalon volume, proportion of ICV, and proportion of TBV
467063
```

```
0.4158783555164566
0.3908427927682916
#this is the Right telencephalon volume, proportion of ICV, and proportion of TBV
470488
0.41892801555727305
0.3937088655790931
```

```
#drop cloumns that are not needed from new df
all_regions_df = kirby906a_ax_df[['type', 'level', 'volume']]


#group by two columns
#"['volume']" tells pandas what to sum by within the subsetted groups
 ##otherwise it sums volume all across
#"reset_index()" binds it back together, it has a weird disjoit look otherwise
all_regions_df = all_regions_df.groupby(['type', 'level'])['volume'].sum().reset_index()
#rename volume to a more descriptive name
all_regions_df=all_regions_df.rename(columns={'volume':'total_volume'})
all_regions_df
```

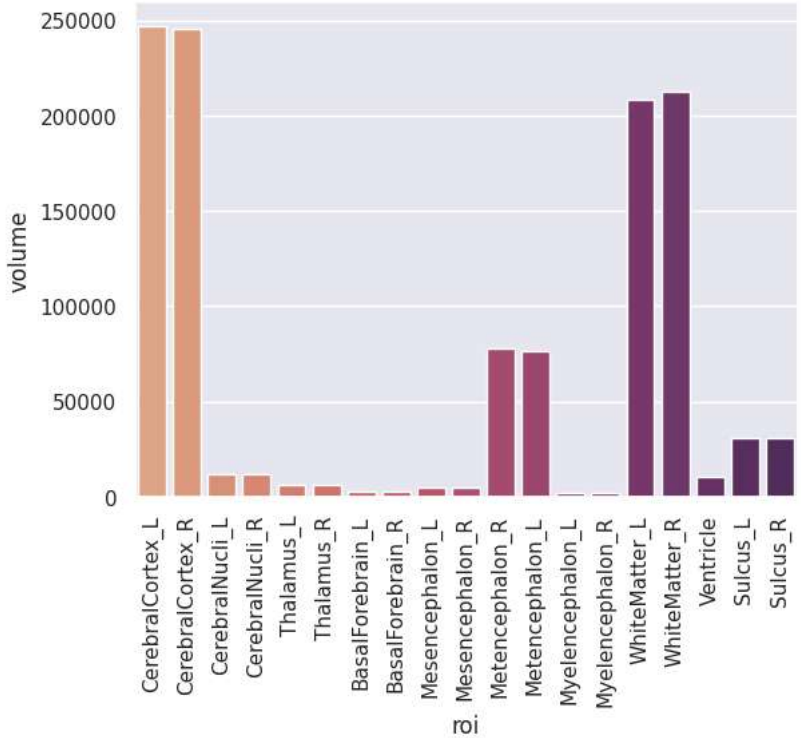
	type	level	total_volume
0	1	1	1195015
1	1	2	1195021
2	1	3	1195034
3	1	4	1195065
4	1	5	1195124
5	2	1	1195015
6	2	2	1195022
7	2	3	1195032
8	2	4	1195041
9	2	5	1195092

Next steps: Generate code with `all_regions_df` New interactive sheet

3. Create a bar plot to visualize this subject's Type 1 Level 2 data using seaborn or matplotlib.
4. Create a donut chart in matplotlib of the Type 1 Level 3 ROI data.
5. Read in the class data from here https://github.com/bcaffo/ds4bme/blob/master/data/classInterests.txt. Name it "class_interests_df". Create a single bar plot of student counts by year and program using seaborn.

```
t1l2 = kirby906a_ax_df[(kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 2)]
bplot = sns.barplot(x="roi", y="volume", data=t1l2, palette="flare")
plt.xticks(rotation=90)
```

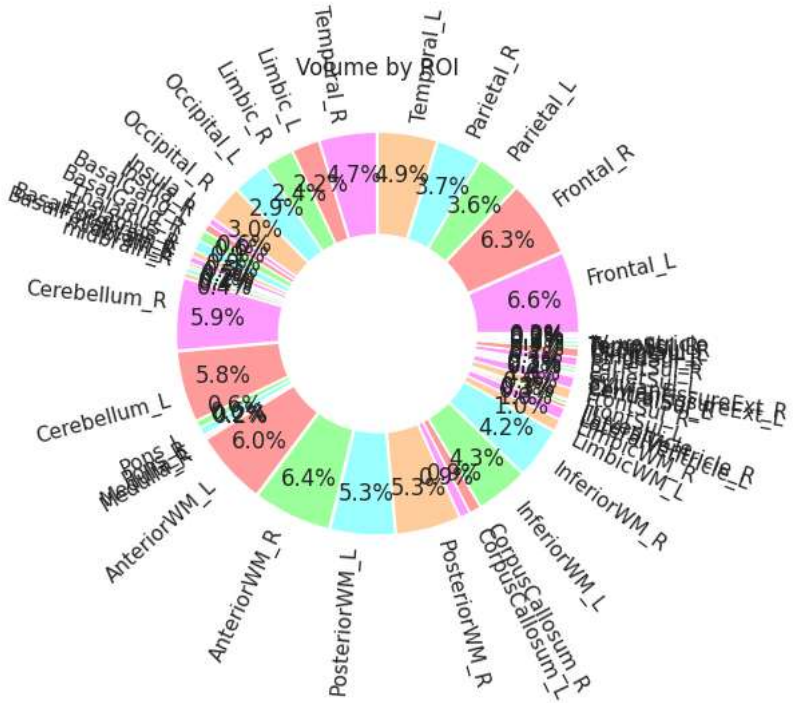
```
/tmp/ipykernel_27771/2121767595.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  bplot = sns.barplot(x="roi", y="volume", data=t1l2, palette="flare")
([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18],
 [Text(0, 0, 'CerebralCortex_L'),
  Text(1, 0, 'CerebralCortex_R'),
  Text(2, 0, 'CerebralNucli_L'),
  Text(3, 0, 'CerebralNucli_R'),
  Text(4, 0, 'Thalamus_L'),
  Text(5, 0, 'Thalamus_R'),
  Text(6, 0, 'BasalForebrain_L'),
  Text(7, 0, 'BasalForebrain_R'),
  Text(8, 0, 'Mesencephalon_L'),
  Text(9, 0, 'Mesencephalon_R'),
  Text(10, 0, 'Metencephalon_R'),
  Text(11, 0, 'Metencephalon_L'),
  Text(12, 0, 'Myelencephalon_L'),
  Text(13, 0, 'Myelencephalon_R'),
  Text(14, 0, 'WhiteMatter_L'),
  Text(15, 0, 'WhiteMatter_R'),
  Text(16, 0, 'Ventricle'),
  Text(17, 0, 'Sulcus_L'),
  Text(18, 0, 'Sulcus_R')])
```




```
#explode = [0.02] * 54
```

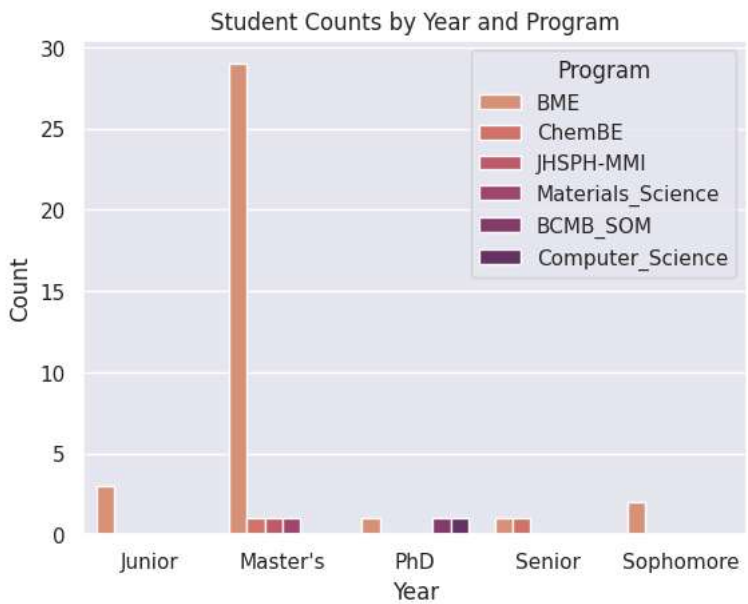
```
t1l3 = kirby906a_ax_df[(kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 3)]


# colors
colors = ['#ff9dff', '#ff9d9d', '#9dff9d',
          '#9dffff', '#ffce9d']
# explosion
explode = [0.02] * 54

# Pie Chart
plt.pie("volume", colors=colors, labels="roi",
        autopct='%1.1f%%', pctdistance=0.79,
        explode=explode, data=t1l3, rotatelabels=True, labeldistance=1.05)

# draw circle
centre_circle = plt.Circle((0, 0), 0.50, fc='white')
fig = plt.gcf()

# Adding Circle in Pie chart
fig.gca().add_artist(centre_circle)

# Adding Title of chart
plt.title('Volume by ROI')
```

```
Text(0.5, 1.0, 'Volume by ROI')
```




```
class_interests_df = pd.read_csv('/content/classInterests.txt')

# Split the 'Year\tProgram' column into 'Year' and 'Program'
```

```
# Split the 'Year\tProgram' column into 'Year' and 'Program'
class_interests_df[['Year', 'Program']] = class_interests_df['Year\tProgram'].str.split('\t', expand=True)

# Count student occurrences by Year and Program size give you the count of rows while naming happens inside the index f
class_interests_df = class_interests_df.groupby(['Year', 'Program']).size().reset_index(name='Count')
sns.barplot(data=class_interests_df, x='Year', y='Count', hue='Program', palette='flare')
plt.title('Student Counts by Year and Program')
plt.xlabel('Year')
plt.ylabel('Count')
plt.show()
```




```
#class_interests_df.head()
```

6. Download the gene expression dataset from here. Name it "gene_expression_df". Information about the data can be found in the sample info here. Do the following operations in order on the gene expression matrix (please note that 2 uses the matrix created in 1 and 3 uses the matrix created in 2):

subtract the rowmean from each row (name it "gene_expression_df_pt1")

take the resulting matrix and subtract the column mean from each column (name it "gene_expression_df_pt2")

Take the resulting matrix and divide each column by the column's standard deviation (name it "gene_expression_df_pt3")

```
gene_expression_df.head()
#len(gene_expression_df)
#gene_expression_df.columns
```

	Unnamed: 0	GSM25581.CEL.gz	GSM25681.CEL.gz	GSM136524.CEL.gz	GSM136707.CEL.gz	GSM25553.CEL.gz	GSM136676.CEL.gz	GSM136711.CEL.gz	GSM136542.CEL.gz	GSM136535.CEL.gz	...
0	1007_s_at	6.333951	5.736190	6.156754	6.513010	6.061013	6.506493	6.863426	6.369645	6.646321	...
1	1053_at	7.255622	7.399993	7.491967	7.006123	7.250995	7.082581	6.842236	7.048487	7.042361	...
2	117_at	5.760106	4.825169	5.039387	5.414160	5.205697	5.300078	5.099337	5.156459	5.736109	...
3	121_at	6.935451	7.025210	7.543667	7.959781	7.223323	8.006816	8.102504	7.434617	7.532321	...
4	1255_g_at	3.597872	3.108801	3.128269	3.364947	3.125909	3.186834	3.311022	3.092042	3.167737	...

5 rows × 209 columns

```
gene_expression_df = pd.read_csv('/content/GSE5859_exprs.csv')
#gene_expression_df
#gene_expression_pt1 = gene_expression_df.groupby("Unnamed: 0").sum().reset_index()
#gene_expression_df_pt1 = gene_expression_df.assign(gene_expression_pt1 = gene_expression_df.sum()/209)
names = gene_expression_df['Unnamed: 0']
gene_expression_df1 = gene_expression_df.drop(columns=['Unnamed: 0'])
gene_expression_df_pt1 = gene_expression_df1.sub(gene_expression_df1.mean(axis=1), axis=0)
```

```
# Subtract the column mean from each column
gene_expression_df_pt2 = gene_expression_df_pt1.sub(gene_expression_df_pt1.mean(axis=0), axis=1)
```

```
# Divide by its standard deviation
gene_expression_df_pt3 = gene_expression_df_pt2.div(gene_expression_df_pt2.std(axis=0), axis=1)

# add names column back
gene_expression_df_pt3.insert(0, gene_expression_df.columns[0], names)
```

7. Download the data from here. Name it "healthcare_df". Create a plot of healthcare spending versus time color coded by states. The option skiprows = 2 is useful in pd.read_csv for this dataset since there's two lines of metadata at the top of the file.
8. Refer to the previous data. Create a barplot using matplotlib of average health care spending by state.

```
healthcare_df = pd.read_csv('https://raw.githubusercontent.com/jhu-advdatasci/2018/master/data/KFF/healthcare-spending.csv', skiprows=2)

# Filter out metadata rows at the end of the dataframe
healthcare_df = healthcare_df[~healthcare_df['Location'].str.contains('For additional analysis', na=False)]
healthcare_df = healthcare_df[~healthcare_df['Location'].str.contains('Note:', na=False)]

# Identify columns that represent years for melting (all columns except 'Location')
id_vars = ['Location']
value_vars = [col for col in healthcare_df.columns if col not in id_vars]

# Melt the DataFrame to a long format suitable for time series plotting
healthcare_df = healthcare_df.melt(
    id_vars=id_vars,
    value_vars=value_vars,
    var_name='Year_Raw', # Temporarily store the original column name
    value_name='Spending'
)
```

```
# Extract the year as an integer from the 'Year_Raw' column
healthcare_df['Year'] = healthcare_df['Year_Raw'].str.extract(r'(\d{4})').astype(int)

# Drop the temporary 'Year_Raw' column as 'Year' now holds the integer year
healthcare_df = healthcare_df.drop(columns=['Year_Raw'])

# Display the head of the transformed dataframe
healthcare_df.head()
```

	Location	Spending	Year
0	United States	675896.0	1991
1	Alabama	10393.0	1991
2	Alaska	1458.0	1991
3	Arizona	9269.0	1991
4	Arkansas	5632.0	1991

Next steps: Generate code with healthcare_df New interactive sheet

```
sns.barplot(data=healthcare_df, x='Year', y='Spending', hue='Location')
```

```
<Axes: xlabel='Year', ylabel='Spending'>
```

For this homework, please return a single ipynb file named hw3.ipynb. Do not use plotly for any of the task. Also, submit a pdf printout of